

Лабораторная работа №9

Настройка POP3/IMAP-сервера

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Цель работы

- Приобрести навыки установки и простой настройки POP3/IMAP-сервера.
- Освоить связку **Postfix + Dovecot** для приёма и доставки почты.
- Научиться проверять работоспособность сервера с помощью клиента и утилит командной строки.

Выполнение работы

```
1 ## Dovecot configuration file
2
3 # If you're in a hurry, see http://wiki2.dovecot.org/QuickConfiguration
4
5 # "doveconf -n" command gives a clean output of the changed settings. Use it
6 # instead of copy&pasting files when posting to the Dovecot mailing list.
7
8 # '#' character and everything after it is treated as comments. Extra spaces
9 # and tabs are ignored. If you want to use either of these explicitly, put the
10 # value inside quotes, eg.: key = "# char and trailing whitespace "
11
12 # Most (but not all) settings can be overridden by different protocols and/or
13 # source/destination IPs by placing the settings inside sections, for example:
14 # protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }
15
16 # Default values are shown for each setting, it's not required to uncomment
17 # those. These are exceptions to this though: No sections (e.g. namespace {})
18 # or plugin settings are added by default, they're listed only as examples.
19 # Paths are also just examples with the real defaults being based on configure
20 # options. The paths listed here are for configure --prefix=/usr
21 # --sysconfdir=/etc --localstatedir=/var
22
23 # Protocols we want to be serving.
24
25 protocols = imap pop3
26
```

```
95
96 # Space separated list of wanted authentication mechanisms:
97 #   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
98 #   gss-spnego
99 # NOTE: See also disable_plaintext_auth setting.
100 auth_mechanisms = plain
101
102 ""
```

Рис. 2: Механизм аутентификации

```
1 # Authentication for system users. Included from 10-auth.conf.
2 #
3 # <doc/wiki/PasswordDatabase.txt>
4 # <doc/wiki/UserDatabase.txt>
5
6 # PAM authentication. Preferred nowadays by most systems.
7 # PAM is typically used with either userdb passwd or userdb static.
8 # REMEMBER: You'll need /etc/pam.d/dovecot file created for PAM
9 # authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
10
11 passdb {
12     driver = pam
13 }
14
15
16 userdb {
17     driver = passwd
18 }
```

Рис. 3: Настройки PAM и passwd

```
15 # There are a few special variables you can use, eg.:
16 #
17 #   %u - username
18 #   %n - user part in user@domain, same as %u if there's no domain
19 #   %d - domain part in user@domain, empty if there's no domain
20 #   %h - home directory
21 #
22 # See doc/wiki/Variables.txt for full list. Some examples:
23 #
24 #   mail_location = maildir:~/Maildir
25 #   mail_location = mbox:~/mail:INBOX=/var/mail/%u
26 #   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
27 #
28 # <doc/wiki/MailLocation.txt>
29 #
30
31 mail_location = maildir:~/Maildir
32
```

Рис. 4: Расположение почтовых ящиков

Настройка межсетевого экрана

```

[root@server.vabazlov.net server]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.vabazlov.net server]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.vabazlov.net server]# firewall-cmd --add-service=imap --permanent
success
[root@server.vabazlov.net server]# firewall-cmd --add-service=imaps --permanent
success
[root@server.vabazlov.net server]# firewall-cmd --reload
success
[root@server.vabazlov.net server]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.vabazlov.net server]# restorecon -vR /etc
[root@server.vabazlov.net server]# systemctl restart postfix
[root@server.vabazlov.net server]# systemctl enable dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' -> '/usr/lib/systemd/system/dovecot.service'.
[root@server.vabazlov.net server]# systemctl start dovecot
[root@server.vabazlov.net server]# █
```

Рис. 5: Настройка межсетевого экрана

This is a summary of the settings which will be used to access your mail.

Account Information

Name:

The above name will be used to identify this account.
Use for example, "Work" or "Personal".

Personal Details

Full Name: vabazlov

Email Address: vabazlov@vabazlov.net

Receiving

Server Type: imapx

Server: mail.vabazlov.net

Username: vabazlov

Security: STARTTLS

Sending

smtp

mail.vabazlov.net

vabazlov

None

Создание тестового письма

The screenshot shows an email composition window titled "test1". At the top left is a blue "Send" button with an arrow icon. To its right is a download icon. The title bar includes a bell icon, an envelope icon, a menu icon, and a close icon. Below the title bar is a toolbar with icons for undo, redo, cut, copy, paste, search, and link. The email fields are as follows: "From:" is set to "vabazlov <vabazlov@vabazlov.net>" with a dropdown arrow; "Signature:" is set to "None" with a dropdown arrow; "To:" is set to "vabazlov@vabazlov.net"; "Cc:" is an empty field; "Subject:" is set to "test1". Below the fields is a formatting bar with a "Plain Text" dropdown, a "Normal" dropdown, and four bulleted list icons. The main text area contains the text "test1" followed by a cursor. At the bottom, there is a "Show Attachment Bar" button, an "Add Attachment..." button, and an "Icon View" dropdown.

Send

test1

undo redo cut copy paste search link

From: vabazlov <vabazlov@vabazlov.net> Signature: None

To: vabazlov@vabazlov.net

Cc:

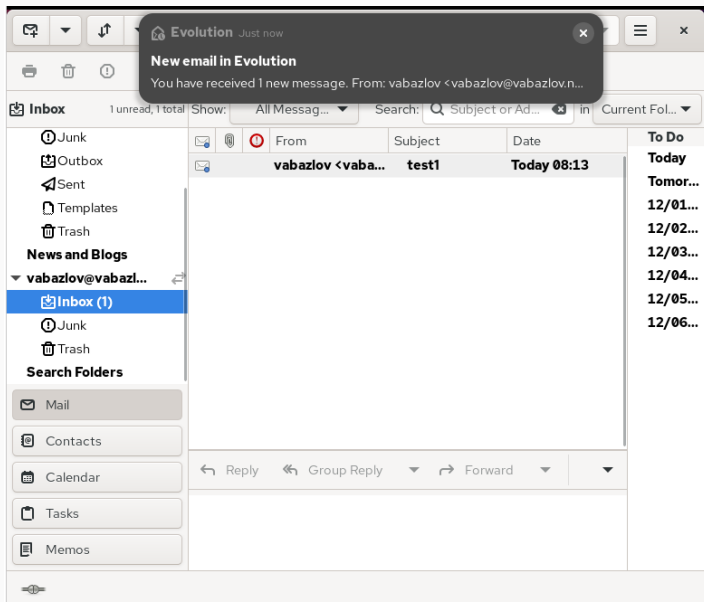
Subject: test1

Plain Text Normal

test1

Show Attachment Bar Add Attachment... Icon View

Полученное письмо во ВХОДЯЩИХ



Записи maillog при работе сервиса

```
[vabazlov@server.vabazlov.net ~]$  
[vabazlov@server.vabazlov.net ~]$ sudo tail -f /var/log/maillog  
[sudo] password for vabazlov:  
Nov 29 08:13:58 server postfix/qmgr[18375]: 1EBBA339F4: removed  
Nov 29 08:14:30 server postfix/smtpd[19350]: connect from client.vabazlov.net[192.168.1.30]  
Nov 29 08:14:30 server postfix/smtpd[19350]: B0B3C339F4: client=client.vabazlov.net[192.168.1.30]  
Nov 29 08:14:30 server postfix/cleanup[19354]: B0B3C339F4: message-id=<7ab95cdbee7eaeae417473142022d8d0228eb.camel@vabazlov.net>  
Nov 29 08:14:30 server postfix/qmgr[18375]: B0B3C339F4: from=<vabazlov@vabazlov.net>, size=535, nrcpt=1 (queue active)  
Nov 29 08:14:30 server postfix/smtpd[19350]: disconnect from client.vabazlov.net[192.168.1.30] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=5  
Nov 29 08:14:30 server postfix/local[19355]: B0B3C339F4: to=<vabazlov@vabazlov.net>, relay=local, delay=0, delays=0/0/0/0, dsn=2.0.0, status=sent  
 (delivered to maildir)  
Nov 29 08:14:30 server postfix/qmgr[18375]: B0B3C339F4: removed  
Nov 29 08:15:40 server dovecot[18619]: pop3-login: Login: user=<vabazlov>, method=PLAIN, rip=192.168.1.1, lip=192.168.1.1, mpid=19567, secured, s  
ession=<RVkOWrdECNjAqAEB>  
Nov 29 08:16:07 server dovecot[18619]: pop3(vabazlov)<19567><RVkOWrdECNjAqAEB>: Disconnected: Logged out top=0/0, retr=1/664, del=1/2, size=1296
```

Рис. 9: Записи maillog при работе сервиса

POP3-сессия через telnet

```
[vabazlov@server.vabazlov.net ~]$  
[vabazlov@server.vabazlov.net ~]$ telnet mail.vabazlov.net 110  
Trying 192.168.1.1...  
Connected to mail.vabazlov.net.  
Escape character is '^]'.  
+OK Dovecot ready.  
user vabazlov  
+OK  
pass 123456  
+OK Logged in.  
list  
+OK 2 messages:  
1 648  
2 648  
.  
retr 1  
+OK 648 octets  
Return-Path: <vabazlov@vabazlov.net>  
X-Original-To: vabazlov@vabazlov.net  
Delivered-To: vabazlov@vabazlov.net  
Received: from client.vabazlov.net (client.vabazlov.net [192.168.1.30])  
by server.vabazlov.net (Postfix) with ESMTP id 1EBBA339F4  
for <vabazlov@vabazlov.net>; Sat, 29 Nov 2025 08:13:58 +0000 (UTC)  
Message-ID: <cec9bffd6dcba9ac1c7132e9e6a618710ece509f.camel@vabazlov.net>  
Subject: test1  
From: vabazlov <vabazlov@vabazlov.net>  
To: vabazlov@vabazlov.net  
Date: Sat, 29 Nov 2025 08:13:58 +0000  
Content-Type: text/plain  
Content-Transfer-Encoding: 7bit  
User-Agent: Evolution 3.52.4 (3.52.4-2.el10_1)  
MIME-Version: 1.0  
  
test1  
.  
dele 2  
+OK Marked to be deleted.  
quit  
+OK Logging out, messages deleted.  
Connection closed by foreign host.  
[vabazlov@server.vabazlov.net ~]$
```

```
connection closed by foreign host.  
[vabazlov@server.vabazlov.net ~]$  
[vabazlov@server.vabazlov.net ~]$ MAIL=~/.Maildir/ mail  
s-nail version v14.9.24. Type '?' for help  
/home/vabazlov/Maildir: 1 message  
▶ 1 vabazlov          2025-11-29 08:13   18/632   "test1  
& q  
Held 1 message in /home/vabazlov/Maildir  
[vabazlov@server.vabazlov.net ~]$
```

Рис. 11: Просмотр Maildir через s-nail

Скрипт provisioning на сервере

```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install postfix
5  dnf -y install s-nail
6  dnf -y install dovecot telnet
7  echo "Copy configuration files"
8  #cp -R /vagrant/provision/server/mail/etc/* /etc
9  echo "Configure firewall"
10 firewall-cmd --add-service=smtp --permanent
11 firewall-cmd --add-service=pop3 --permanent
12 firewall-cmd --add-service=pop3s --permanent
13 firewall-cmd --add-service=imap --permanent
14 firewall-cmd --add-service=imaps --permanent
15 firewall-cmd --reload
16 restorecon -vR /etc
17 echo "Start postfix service"
18 systemctl enable postfix
19 systemctl start postfix
20 echo "Configure postfix"
21 postconf -e 'mydomain = user.net'
22 postconf -e 'myorigin = $mydomain'
23 postconf -e 'inet_protocols = ipv4'
24 postconf -e 'inet_interfaces = all'
25 postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
26 postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
27 postconf -e 'home_mailbox = Maildir/'
28 postfix set-permissions
29 restorecon -vR /etc
30 systemctl stop postfix
31 systemctl start postfix
32 systemctl restart postfix
33 systemctl enable dovecot
34 systemctl start dovecot
35
36
```



```
1  #!/bin/bash
2  echo "Provisioning script $0"
3  echo "Install needed packages"
4  dnf -y install postfix
5  dnf -y install s-nail
6  dnf -y install evolution
7
8  echo "Configure postfix"
9  postconf -e 'inet_protocols = ipv4'
10 echo "Start postfix service"
11 systemctl enable postfix
12 systemctl start postfix
```

Рис. 13: Скрипт provisioning на клиенте

Итоги работы

- Развёрнут и настроен почтовый сервер **Postfix + Dovecot**.
- Реализована работа по протоколам **POP3** и **IMAP**.
- Подтверждена корректная доставка и чтение почты через Evolution, telnet и s-nail.
- Настроен журнал маил-сервера и проанализированы основные события.
- Подготовлены скрипты provisioning для автоматического развёртывания стенда.